

Start

every plan zero but the household load's;
price zero (or a warm start)

One iteration, repeated

Every terminal, at the same time

its cheapest plan near a target: its last
plan, moved by the imbalance and the price
devices: their proximal step, by any solver
(a DP; an LP for a battery), no tariff; the
tank and HVAC relaxed
grid connection: the bill and the limits, in
closed form

plans

imbalance, price

The house

average the imbalance; raise the price
where it persists
make the plan runnable (tank, HVAC
on/off) and keep the best so far

the plans settle, or 100
iterations

Result

the best runnable plan, then a polish:
devices re-plan one at a time, kept only if
cheaper
*no bound: how far from the best is
unknown*